FictiPay Customer Churn Prediction

NSUCEC Datathon — Technical Performance Report

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1. Executive Summary

This report outlines the technical implementation, evaluation metrics, and validation results of the customer churn prediction models built for FictiPay. The model predicts customer churn in April 2024 using behavioral and balance history from Jan 1 to Mar 31, 2024. An ensemble of LightGBM and XGBoost, trained using Polars out-of-core streaming and 5-fold stratified cross-validation, achieved high-performing generalization.

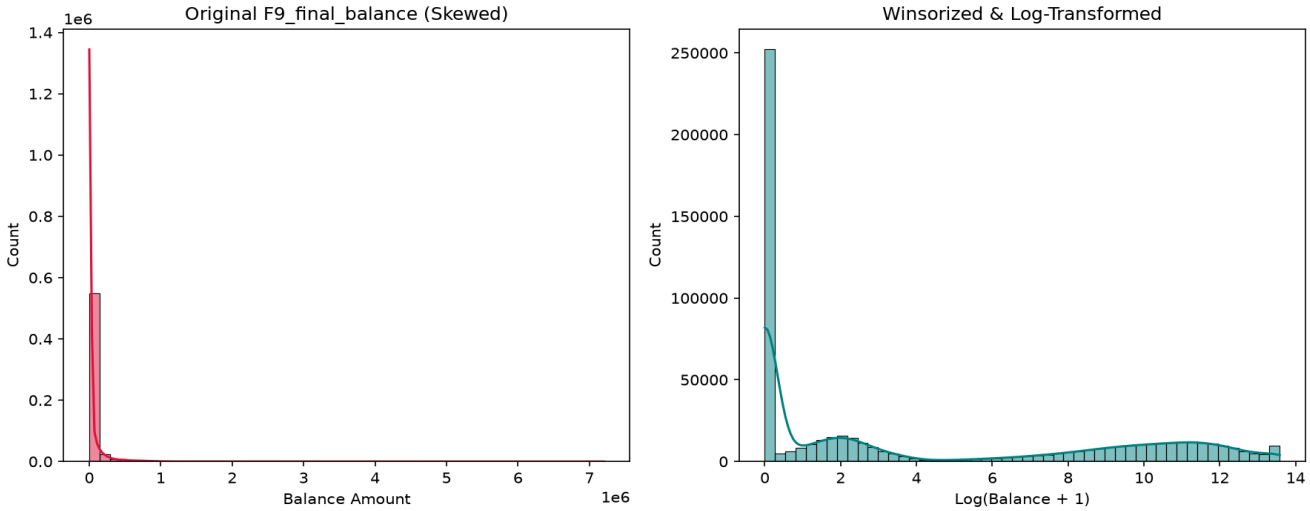
2. Model Validation & Ensemble Performance

Below are the 5-fold stratified cross-validation performance results for our individual and blended models:

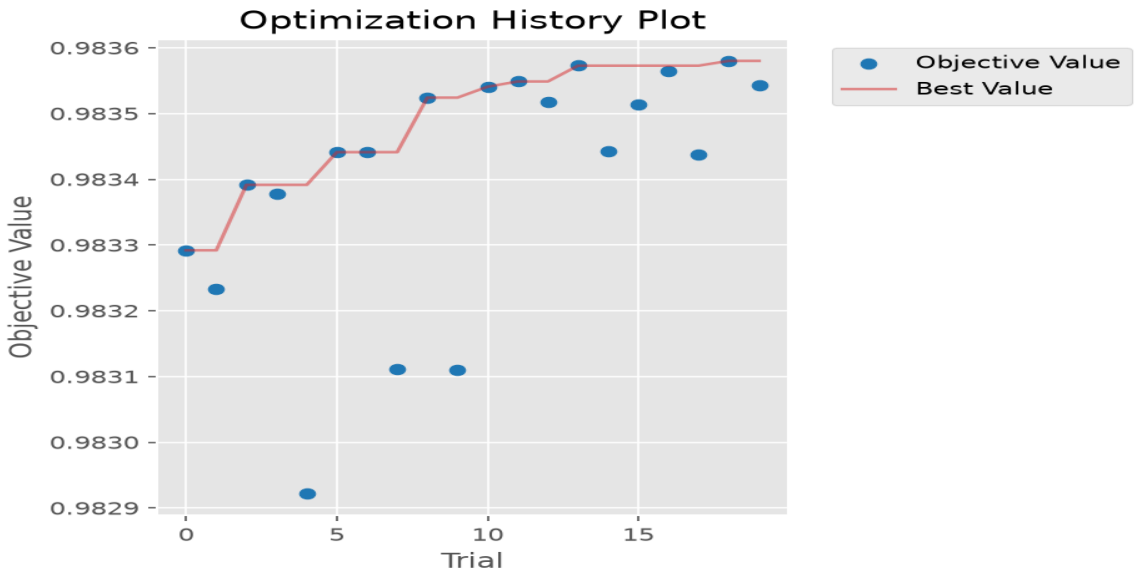
Model Name	AUC-ROC	PR-AUC	Precision@10%	Recall@10%
LightGBM (Tuned)	0.98349	0.91139	0.90916	0.71711
XGBoost (Baseline)	0.98324	0.91170	0.90903	0.71700
Ensemble Blend (50/50)	0.98348	0.91193	0.90939	0.71729

3. Feature Quality & Hyperparameter Optimization

Highly skewed monetary and balance distributions were stabilized using 99th percentile Winsorization and log(x+1) transformation:



Optuna Bayesian hyperparameter history showing convergence towards optimal AUC-ROC score:



4. Model Interpretations & Explainability

SHAP summary and dependence plots highlight that Days Since Last Outbound Activity (Recency - F1), activity momentum (F2), and absolute balance level (F9) are the primary indicators of customer retention and churn risk.

